

Track DC v0.10/v0.11/v0.13 — K59-analog at chain levels rank/ N_c /g consolidated

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Track DC v0.10/v0.11/v0.13 — chain levels consolidated

1. Setup

Per Keeper menu #6-#9: K59-analog substrate-operator connection at each chain level. v0.12 ($X=n_C$) already filed; this doc consolidates v0.10 ($X=\text{rank}$), v0.11 ($X=N_c$), v0.13 ($X=g$) into one framework.

2. v0.10 — $X=\text{rank}$ (seed level)

2.1 Setup

$X = \text{rank} = 2$; $\text{GF}(2^{\text{rank}}) = \text{GF}(4)$; $M_2 = 3 = N_c$ (Mersenne prime).

2.2 Substrate-operator connection candidates

- 2-step cyclotomic chain on $\text{GF}(4)$ (rank steps)
- Reed-Solomon at $\text{GF}(4)$: block length $3 = N_c$ (substrate-natural)
- Substrate-operator action: smallest finite-field seed for substrate computation

2.3 Multi-week derivation

Explicit 2-step chain on $\text{GF}(4)$; RS coding capacity at small block length; substrate-tick at seed-level.

3. v0.11 — $X=N_c$ (color-level)

3.1 Setup

$X = N_c = 3$; $\text{GF}(2^{N_c}) = \text{GF}(8)$; $M_3 = 7 = g$ (Mersenne prime).

3.2 Substrate-operator connection candidates

- 3-step cyclotomic chain on $\text{GF}(8)$ (N_c steps)
- Reed-Solomon at $\text{GF}(8)$: block length $7 = g$; cyclic-prime structure
- Substrate-operator action: color $\text{SU}(3)$ -related computational tier

3.3 Connection to color $\text{SU}(3)$

$\text{GF}(8)$ has multiplicative-cycle $= 7 = g$. Substrate's color-tier finite field has cycle equal to outer-tier exponent.

Substrate-mechanism reading: color $\text{SU}(3)$ substrate computation operates at $\text{GF}(8)$ substructure level; multi-week verification.

4. v0.13 — $X=g$ (outer-tier RATIFIED)

4.1 Setup

$X = g = 7$; $GF(2^g) = GF(128)$; $M_7 = 127$ (Mersenne prime). K59 RATIFIED.

4.2 Substrate-operator action — formalization

Per K59 RATIFIED + Paper #122: - 7-step cyclotomic chain on $GF(128)$ - Reed-Solomon at $GF(128)$ with block length 127 ($= M_g$); code dimension up to $7 = g$ - Substrate-operator action via $GF(128)$ field arithmetic - $\hat{T}_{\text{tick}}$: 7-step cyclotomic per Koons tick

4.3 Bergman boundary connection

At outer-tier: substrate's Shilov boundary projection uses $GF(128)$ field structure. Bergman kernel evaluation via Reed-Solomon syndrome.

4.4 RATIFIED status

K59 at $X=g$ is RATIFIED Casey-named principle. v0.13 documents the substrate-operator connection at this level for cross-CI reference + Multi-phase quiver v0.2+ Hall algebra construction.

5. Cross-level structure summary

Level X	$GF(2^X)$	Steps	M_X	Substrate role
rank=2	$GF(4)$	2	$3=N_c$	Seed-level (v0.10)
$N_c=3$	$GF(8)$	3	$7=g$	Color-level (v0.11)
$n_C=5$	$GF(32)$	5	31	Inner-tier (v0.12)
$g=7$	$GF(128)$	7	127	Outer-tier RATIFIED K59 (v0.13)

Self-referential Mersenne nesting (per v0.7 Section 4.5): $GF(4)$'s $M_2=3=N_c$; $GF(8)$'s $M_3=7=g$; substrate's smallest levels have multiplicative-cycle structure pointing at substrate's higher chain primaries.

6. Multi-week derivation path

- v0.14 explicit per-level RS code capacity verification (multi-week)
- v0.15 cross-tier coupling mechanism (multi-week)
- v0.16 substrate-tick scaling per chain level (multi-week)
- v0.17 connection to Multi-phase quiver Hall algebra structure (multi-week)

7. Honest scope

What's RATIFIED: - K59 at $X=g$ (RATIFIED Casey-named) - Standard $GF(2^X)$ finite-field theory at primes - A_{sub} v0.9 super-quiver

What's FRAMEWORK in v0.10-v0.13: - Substrate-operator connection candidates per chain level - Self-referential Mersenne nesting structural reading

What's INTERPRETIVE / OPEN (Cal #29 STANDING): - Cross-tier coupling mechanism - Substrate-tick scaling per tier - Specific RS coding capacity per level

— Lyra, Track DC v0.10/v0.11/v0.13 chain levels consolidated v0.1 filed Wednesday 2026-05-27 ~11:12 EDT. FRAMEWORK. Complete chain-level K59-analog substrate-operator connection framework set; multi-week explicit derivation per level continues.